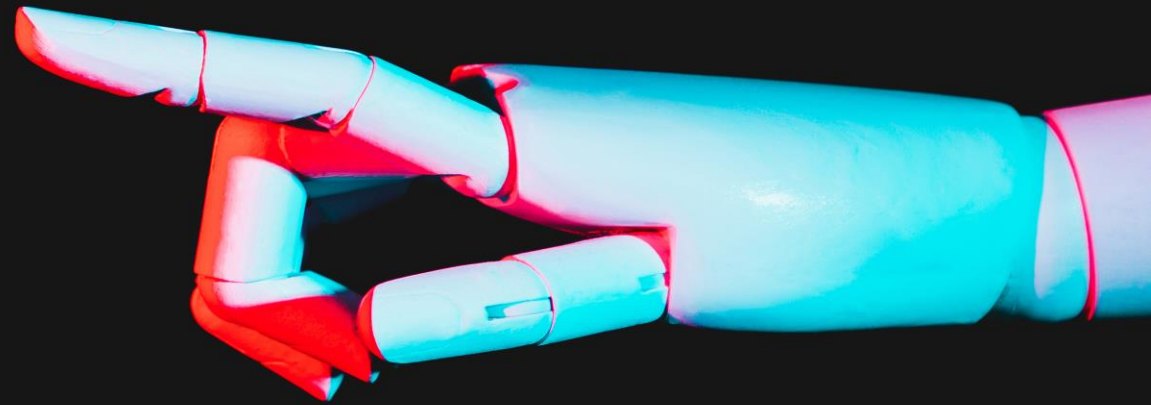
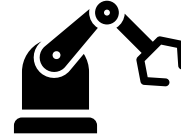


Machine Learning 600

Logistic Regression | KNN | Decision Tree | PCA



Problem



- Financial institutions need a better way to identify high-risk and low-risk loan applicants.
- This project uses supervised machine learning to predict loan status.
- The aim is to reduce false approvals and false rejections.

Dataset Overview

Dataset: 105 rows and 12 columns.

Target variable: loan_status.

0 means lower risk and 1 means higher risk/default.

Features include: age, income, home ownership, loan amount, interest rate and credit history.

Data Issues Identified

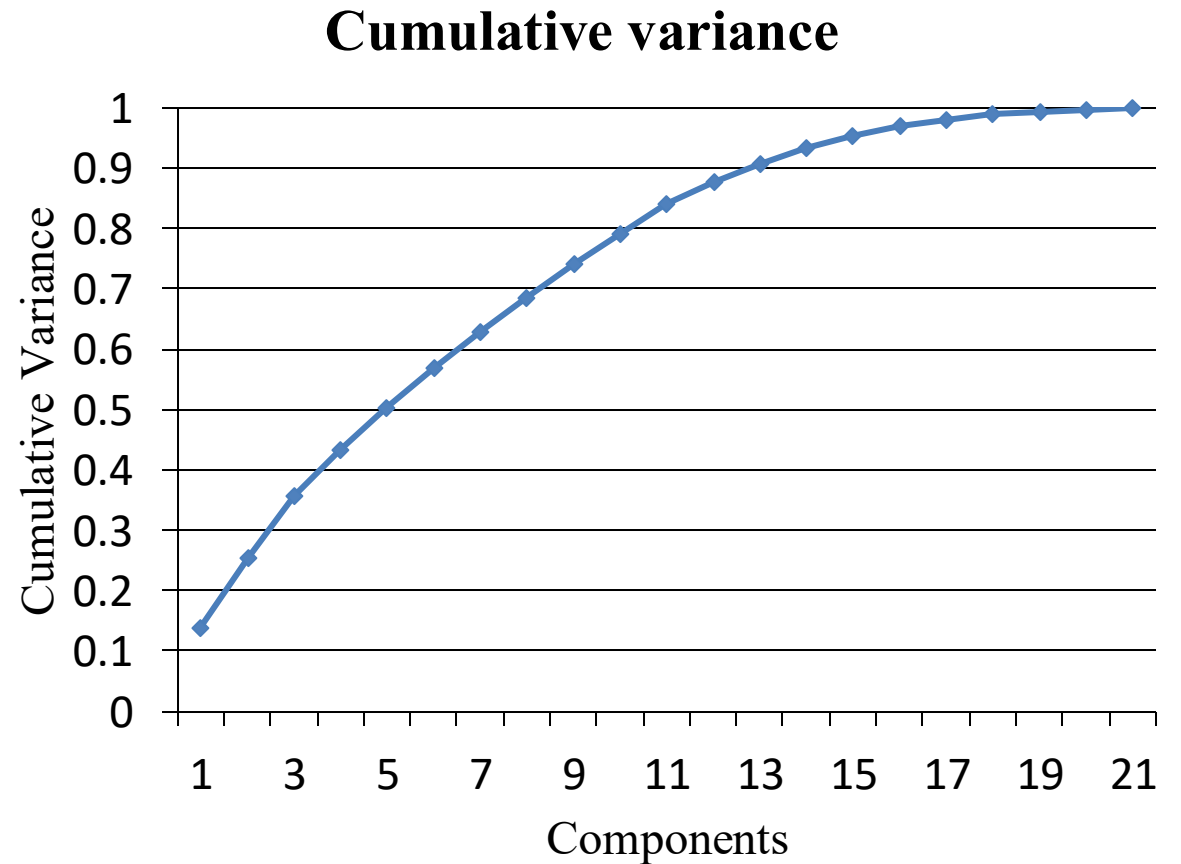
- **Missing values found: 12.**
- **Duplicate rows found: 0.**
- Some numerical features contain outliers.
- Categorical data must be converted into numbers before modelling.

Preprocessing Summary

- ✓ Removed duplicate rows.
- ✓ Filled missing numerical values with the median.
- ✓ Filled missing categorical values with the mode.
- ✓ Capped outliers using the IQR method.
- ✓ Used one-hot encoding, feature scaling and a 80/20 train-test split.

PCA

- PCA reduces the number of features.
- It keeps most of the useful information.
- 15 components keep at least 95% of the variance.
- This can make the dataset simpler before modelling.



Model Comparison - Before PCA

Model	Accuracy	Precision	Recall	F1
Logistic Regression	0.86	1.00	0.77	0.87
KNN	0.67	0.71	0.77	0.74
Decision Tree	0.81	0.91	0.77	0.83

Model Comparison - After PCA

Model	Accuracy	Precision	Recall	F1
Logistic Regression	0.86	1.00	0.77	0.87
KNN	0.71	0.77	0.77	0.77
Decision Tree	0.90	1.00	0.85	0.92

Hyperparameter Tuning

GridSearchCV was used to tune the Decision Tree.

Best parameters: {'max_depth': 2, 'min_samples_split': 2}.

Tuned Decision Tree test accuracy: 0.81.

Tuning helps control model complexity and can reduce overfitting.

Conclusion and Insights

- Logistic Regression had the highest accuracy before PCA.
- PCA reduced the number of features while keeping most of the variance.
- The model can support faster and more consistent credit decisions.
- False approvals may cause financial loss and false rejections may cause lost customers.
- The final model should be tested on the complete dataset before real business use.